

## TEAM CONTRIBUTION RECORD

**Team size: 4 members | Duration: 5 days × 10 hours/day (50 hours per member, 200 hours total)**

| Member              | Role                                    | Area of Ownership                                                                                                                                                                                         |
|---------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Châu Quốc Lâm Phong | AI & Ingestion Engineer                 | Document ingestion (PDF/DOCX reading, chunking, validation), Gemini API integration, prompt templates and prompt versioning                                                                               |
| Đoàn Thị Quỳnh Nhi  | Backend & Rule Engine Engineer          | FastAPI backend skeleton, Pydantic schemas, the pure-Python rule engine (coverage scoring, prerequisite checking), comparison/hallucination/contradiction detection, prompt-injection security filtering  |
| Phạm Tấn Tài        | Fullstack & Database Developer          | Database schema and ORM models, seed data, all user-facing screens (document ingestion, plan detail, reviewer dashboard, learner portal, quiz view), Docker packaging and deployment                      |
| Lê Thị Kiều Duyên   | QA, Security, Data & Documentation Lead | Sample and adversarial test documents, the Role Requirement Matrix source data, adversarial/prompt-injection test scenarios, project documentation, technical blog, final report and submission checklist |

### Contribution by Phase

Day 1 - Environment, Foundational Data & Database

| Member       | Contribution                                                                                                                                                                                                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phong</b> | Set up the Python environment and .env; built the PDF/DOCX readers; wrote the chunking module (heading/section-based, with stable chunk and section ids); wrote the file-validation module (format, size, empty-file checks); wrote unit tests for ingestion and validation |
| <b>Nhi</b>   | Built the FastAPI skeleton with CORS and a health-check endpoint; defined the core Pydantic schemas (onboarding plan, module, task, quiz, document chunk); built the initial upload endpoint                                                                                |
| <b>Tài</b>   | Configured the database connection; created the ORM models and ran the initial migration; seeded demo accounts (admin, reviewer, employee)                                                                                                                                  |
| <b>Duyên</b> | Authored 10 realistic sample policy documents (HR, IT security, finance compliance, onboarding); built the initial Role Requirement Matrix covering 10 job positions                                                                                                        |

Day 2 — Dual-Pipeline Core

| Member       | Contribution                                                                                                                                                                                                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phong</b> | Built the Gemini API client with retry/backoff; designed the structured-output response schemas (plan → module → task → quiz, each carrying a source citation); wrote the v1.0 prompt template for onboarding-plan generation; added prompt-version tracking to generated output |
| <b>Nhi</b>   | Wrote the pure-Python Coverage Scorer (role-matrix match percentage, no AI dependency) and the Prerequisite Checker (module ordering rules); wrote unit tests for both                                                                                                           |
| <b>Tài</b>   | Built the document-ingestion UI (drag-and-drop upload, progress bar, chunk-list display); built the plan-detail UI (modules/tasks as an expandable accordion)                                                                                                                    |
| <b>Duyên</b> | Expanded the sample-document set to 20 files across formats; wrote the initial project report outline                                                                                                                                                                            |

### Day 3 - Comparison Engine & Reviewer Interface

| Member       | Contribution                                                                                                                                                                                                                                                                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phong</b> | <i>(joint with Nhi)</i> Defined the shared data contract between the two pipelines; built the Comparison Engine (field-by-field diff between Pipeline 1 and Pipeline 2 output); built the Hallucination Detector; built the final-status classifier (Verified / Verified with Warning / Manual Review Required); built the Contradiction Checker |
| <b>Nhi</b>   | <i>(joint with Phong)</i> Same task set as above — comparison engine, hallucination detection, status classification, and contradiction checking were built collaboratively                                                                                                                                                                      |
| <b>Tài</b>   | Built the dual-comparison reviewer UI (Pipeline 1 vs. Pipeline 2 side by side, with mismatch/warning highlighting); wired Approve/Reject/Override actions to write complete audit-log entries                                                                                                                                                    |
| <b>Duyên</b> | Wrote 10 prompt-injection test scenarios ("ignore previous instructions," "system override," "DAN mode," etc.); prepared adversarial sample documents with expired or internally contradictory policies                                                                                                                                          |

### Day 4 — Hidden-Test Readiness, Blog & Demo Video

| Member       | Contribution                                                                                                                                                                                                                                   |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phong</b> | Ran the hidden-test scenario against a completely unseen document end to end; hardened the pipeline against edge cases (empty files, scanned PDFs with no text layer, UTF-16 encoding, oversized uploads); wrote the demo script and shot list |
| <b>Nhi</b>   | Supported hidden-test verification of the rule engine against unseen data; wrote the prompt-injection filtering middleware that screens text before it reaches Gemini; supported demo-video production                                         |
| <b>Tài</b>   | Built the learner portal (module/task checklist with persisted completion state); built the quiz-taking screen (source-cited questions, immediate scoring on submit)                                                                           |

| Member       | Contribution                                                                                                                                                                                                   |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Duyên</b> | Wrote and published the required 2,000+ word technical blog covering system architecture, the dual-pipeline design, and results; supported demo-video prep and ran an initial pass of the submission checklist |

### Day 5 — Deployment Packaging & Handoff

| Member       | Contribution                                                                                                                                                                                                                                                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phong</b> | <i>(joint with Nhi)</i> Cleaned up the codebase against team coding rules (no stray comments, no TODOs, no bare except Exception: pass); wrote the complete README.md; audited git history for accidentally committed secrets; ran the full test suite and fixed remaining bugs; created the final commit and v1.0.0 release tag |
| <b>Nhi</b>   | <i>(joint with Phong)</i> Same task set as above — code quality, README, history audit, test-suite pass, and release tagging were completed collaboratively                                                                                                                                                                      |
| <b>Tài</b>   | Wrote the Dockerfile and docker-compose.yml for the full stack; deployed to the cloud host with production environment variables; ran production smoke tests across all major user flows                                                                                                                                         |
| <b>Duyên</b> | Finalized the project report (architecture diagrams, SRS mapping, screenshots, test results, conclusion); verified all 18 submission-checklist items; submitted the final entry (GitHub link, deployed URL, video, blog, report)                                                                                                 |

### Team Working Principles (from the WBS)

- Commit frequency: at least 1–2 commits per person per day, on that member's own branch.
- Merge process: every feature branch is opened as a Pull Request and reviewed by another member before merging to main.
- Secrets: .env files, API keys and database passwords are never committed - enforced by keeping .env.example as the only tracked template.
- AI-usage transparency: every AI-assisted code or documentation contribution is logged in AI\_USAGE.md.
- Code quality: comments explain *why*, not *what*, per the team's own coding rules - no "what this does" comments, no leftover TODOs, no bare except Exception: pass.